

Patterns of spring-freeze risk for temperate trees contributes to phenological cue differences, but leaves much unexplained

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Abstract

Phenological timing in temperate woody plants balances the advantage of early leafout with the risk of late spring freezes, influenced by environmental cues such as forcing temperatures, chilling, and photoperiod. The spring predictability hypothesis (SPH) suggests species in less predictable spring climates rely more on chilling and photoperiod to avoid freeze damage, but this has not been rigorously tested. Here we analyzed species-level phenological cue sensitivities across temperate North American and European woody plants using Bayesian hierarchical models and climate data from species' ranges. We found limited intercontinental differences and mixed support for SPH predictions, with some associations between spring predictability and chilling sensitivity but inconsistent patterns overall. These results indicate that while spring predictability influences phenological cues, other ecological and evolutionary factors also shape cue responses. Understanding these complexities is essential for predicting phenological shifts under climate change.

Introduction

Phenology, the timing of annual life cycle events, allows for organisms to match critical life-cycle transitions with optimal environmental conditions. Through the phenology of spring budburst, temperate woody plants balance the resource and competition advantages of precocious leafout with the risk of damage from late season freezes (Savage and Cavender-Bares,

2013). To navigate this trade-off, woody plants have evolved complex physiological responses to sense environmental cues that signal the arrival of suitable conditions for resuming growth (Polgar and Primack, 2011). Decades of research on phenology suggest that warming spring temperatures (forcing), cool winter temperatures (chilling) and day length (photoperiod) are the primary environmental cues for temperate woody plant phenology (Ettinger et al., 2020; Forrest and Miller-Rushing, 2010). These studies also demonstrate that there are substantial differences among species in their reliance on particular cues than others (Laube et al., 2014; Pau et al., 2011). Yet, our understanding of why species differ in their cue responses remains limited, and better knowledge of the ecological and evolutionary drivers shaping phenological cues is critical for predicting the magnitude and impacts of phenological shifts with climate change.

The predictability of the arrival of spring may strongly influence the evolution of phenological cues (Zohner et al., 2017; Zohner and Renner, 2017; Dawson-Glass et al., 2025). In regions where the start of spring is unpredictable, species are expected to evolve a stronger dependence on chilling and photoperiod cues to prevent premature leafout and reduce the risk of freeze damage. Conversely, in regions where the seasonal warming reliably indicates the start of spring, species should respond strongly to forcing, with less reliance on chilling or photoperiod. This concept, known as the spring predictability hypothesis (SPH), is intuitive and has received some recent support in the literature (Zohner et al., 2017). However, the SPH assumes that species' phenological responses are at a stable equilibrium with their environment, which may not always be the case (Normand et al., 2011; Lechowicz, 1984). It is also unclear the time scale at which season predictability would shape cue responses. Spring predictability could drive selective pressure to increase chilling and/or photoperiod sensitivity on an evolutionary time scale, or define species ranges based on their inherited cue sensitivities on an ecological time scale. Testing the predictions of SPH across multiple geographic scales can serve to evaluate this hypothesis, and offer an improved understanding of the drivers of biogeographic patterns of phenology cue sensitivity.

Spatial predictions of the SPH

Global circulation patterns generate substantially different spring climatic conditions on either side of the North Atlantic (). In Eastern North America, spring is marked by climatic instability, while in Europe, the arrival of spring is generally more consistent (Figure 1). Given these contrasting climate regimes, the SPH predicts that North American species should exhibit stronger sensitivity to chilling and photoperiod cues and weaker sensitivity

to forcing (Dawson-Glass et al., 2025). We tested these predictions by comparing the cue sensitivities of North American and European species estimated from Bayesian hierarchical models.

It is also possible that phenological responses to cues play a significant role in determining species' range limits (Chuine, 2010). Species that rely primarily on forcing cues should have distributions restricted to geographic regions where spring predictability is high, whereas species that rely more heavily on chilling and photoperiod cues can persist in regions with lower spring predictability. If this is the case, the SPH predicts that within each continent, species with higher chilling and photoperiod cue responses should be associated with lower spring predictability across their ranges. We tested this prediction by regressing the species-level posterior estimates of forcing, chilling and photoperiod sensitivity from our previously described model against two metrics of spring predictability, with separate models for each cue by continent.

Alternatively, local adaptation in phenological cue sensitivity may overwhelm any relationship between species-level cue use and range-wide climate conditions. However, if cues are locally adapted, then neither continental differences or range wide climate conditions would be strongly associated with cue use. To assess this possibility, we designed a two-level, hierarchical model using data from studies of woody plant budburst that sampled species from multiple provenance locations, allowing us to partition variance between populations and species.

Our study interrogated the SPH; one of the most fundamental biological assumptions relating to the evolution of phenological cue responses that is frequently asserted but rarely tested (but see Zohner et al., 2017). Our analyses found only mixed support for the SPH, highlighting the need to more holistically integrate additional drivers, such as community and phylogenetic factors, into frameworks for understanding the evolution of phenological cues.

Materials and Methods:

Species' range characteristics and climate data

We extracted climate data from daily gridded meteorological datasets for both Europe and North America. For Europe, we extracted minimum and maximum daily temperatures from the E-OBS dataset (<https://cds.climate.copernicus.eu/cdsapp#!/dataset/>

insitu-gridded-observations-europe?tab=overview; last accessed on October 2021) corresponding to the period comprised between 1980 and 2016. Specifically, we used version 17 at a resolution of 0.5 latitudinal degrees. For North America, we extracted minimum and maximum daily temperatures from Justin Sheffield’s Princeton Global Forcing dataset (<http://hydrology.princeton.edu/data/pgf/v3/0.25deg/daily/>; last accessed on October 2021) for the same period. We used version v3 at a resolution of 0.25 latitudinal degrees.

For 22 European and 16 North American tree species, we obtained published distributional range maps in shapefile format. European species ranges were downloaded from (<http://www.sciencedirect.com/science/article/pii/S2352340917301981?via%3Dihub#ec-research-data>; last accessed on XXX) (Caudullo et al., 2017) and North American ranges were obtained from <https://www.fs.fed.us/nrs/atlas/littlefia/#> (Prasad, 2003). For each species’ range, we extracted climate data corresponding to all grid cells contained within the range.

We used minimum and maximum daily temperatures within species’ ranges and computed Growing Degree Days (GDD), Growing Degree Days until the last frost and Spring Temperature Variability (STV). GDD was calculated as the summed temperatures above 10C recorded from January 1st until May 31st. GDD to the last frost was calculated as GDD but instead of summing temperatures above a threshold until a fixed date, the sum was performed until the date at which the latest minimum temperature below -5C was recorded. We then calculated the standard deviation (hereafter: Stdv in GDDs to last frost) to this date as one proxy for spring predictability. We then calculated STV, the standard deviation of mean minimum temperature from March 1st until May 31st (Zohner et al., 2017). Specifically, we computed these metrics variability within each location across years (1980 to 2016) and for each year across the grid cells comprised within each species’ range.

Statistical analyses

To estimate species-level responses to forcing, chilling and photoperiod, we analyzed phenological observations from the Observed Spring Phenology Responses in Experimental Environments (OSPREE database (Wolkovich et al., 2019)) using a Bayesian Hierarchical model developed by Morales-Castilla et al. (2024). This model estimates forcing, chilling and photoperiod sensitivities through phylogenetically-structured partially pooling. The model was fit using Stan modeling language (?) (www.mc-stan.org, and the data and code used to run this model are available at X. From this model, we extracted the species-levels posterior

estimates of forcing, chilling and photoperiod sensitivity and tested whether European and North American taxa had systematically different cue sensitivities by comparing the cues using linear regression, with native continent as a main effect predictor.

To assess the relationship between spring predictability—estimated across species’ ranges—and cue sensitivity, we regressed the posterior estimates of cue sensitivity against spring predictability. We used separate models for European and North American taxa, and modeled each cue sensitivity (forcing, chilling and photoperiod) separately. We also ran separate models using STV and Stdv in GDDs to last frost as proxies for spring predictability, as main effects. We ran these model using the R package BRMS. (CITES, version X).

Finally, to assess variation within and across sites, we designed a two-level, hierarchical model using data from the OSPREE database. We subsetting the studies to include only those that had multiple provenance locations. Since chilling estimates were either from experimental chilling, from field chilling or a combination of both, we removed ‘chilling’ as a predictor for this model as it correlated strongly with provenance latitude and would result in nonidentifiability in our results. We used a Bayesian mixed-effects hierarchical model approach to analyze our data to best estimate the day of budburst. This model fit a Gaussian distribution using study, species and population as intercepts, forcing and photoperiod as predictors (fixed effects) and species nested within population (i.e., site) as modeled groups (random effects). The Bayesian model was fit using Stan modeling language (www.mc-stan.org), accessed via the *rstan* package (version 2.15.1), version 2.3.1, in R (<https://www.r-project.org/>), version 3.3.1. We ran four chains, with 2,500 warm-up iterations followed by 3,000 sampling iterations, resulting in 12,000 posterior samples for each parameter.

For all models, we assessed good model performance by ensuring $\hat{R}$ were close to 1, n_{eff} values were high, and by visually inspecting chain convergence and posterior distributions (Gelman and Hill, 1997).

Results

Inter-continental Climatic Pattern

We found no substantial differences in cue sensitivity between continents (Figure: 2). Mean forcing sensitivity for European species was $-6.76 \text{ } UI_{95}(-17.80, 2.08)$ and $-7.94 \text{ } UI_{95}[-17.90, 1.93]$ for North American species. Mean chilling sensitivity for European species was $-8.44 \text{ } UI_{95}(-22.60, 4.69)$

and $-8.76 \text{ } UI_{95}[-26.90, 4.82]$ for North American species. Mean photoperiod sensitivity for European species was $-1.36 \text{ } UI_{95}[-5.91, 2.89]$ and $-1.35 \text{ } UI_{95}[-5.88, 2.98]$ for North American species.

Within continent ranges

We found mixed support for the continent level predictions of the SPH. With STV as the metric for spring predictability, we found that in Europe spring predictability across a species' range had no clear relationship to forcing or photoperiod cues (mean forcing effect: $-0.59 \text{ } UI_{95}[-4.44, 3.01]$, mean photoperiod effect: $1.34 \text{ } UI_{95}[-0.49, 3.24]$). Contrary to the predictor of the SPH, decreasing spring predictability was associated with weaker chilling sensitivity (mean chilling effect: $6.37 \text{ } UI_{95}[3.99, 7.97]$).

For North American species, spring predictability also had no clear relationship to forcing or photoperiod cue sensitivity (mean forcing effect: $-0.05 \text{ } UI_{95}[-2.26, 2.06]$, mean photoperiod effect: $0.15 \text{ } UI_{95}[-0.93, 1.21]$). For chilling, decreasing spring predictability was associated with increased chilling sensitivity (mean chilling effect: $-3.45 \text{ } UI_{95}[-7.50, 0.69]$), supporting the prediction of the SPH.

With standard deviation in growing degree days to last frost as the metric for spring predictability, we observed different relationships between spring predictability and species-level cue sensitivity than we found using STV. For European species, we found decreasing spring predictability was associated with both higher forcing sensitivity (mean forcing effect: $-3.61 \text{ } UI_{95}[-9.04, -1.64]$) and chilling sensitivity (mean chilling effect: $-5.27 \text{ } UI_{95}[-11.45, -1.36]$), while there was no clear effect of spring predictability on photoperiod sensitivity (mean photoperiod sensitivity: $-0.80 \text{ } UI_{95}[-3.62, 2.09]$).

For North American species, decreased predictability was associated with weaker forcing sensitivity though the strength of this association was relatively weak (mean forcing effect: $0.09 \text{ } UI_{95}[-0.04, 0.24]$). There was no clear relationship between spring predictability and chilling sensitivity (mean chilling effect: $-0.02 \text{ } UI_{95}[-0.34, 0.31]$). For photoperiod sensitivity, we observed a moderate association between decreasing spring predictability and photoperiod sensitivity (mean photoperiod effect: $-0.04 \text{ } UI_{95}[-0.11, 0.04]$).

Intra-and Interspecific cue variation

We detected limited population-level variation in forcing and photoperiod cue sensitivity, with within-species variation being less substantial than among-species variation (Fig. 6). Notably, we found the largest source of variation in phenological cue estimates was attributable to the study effect (Fig. 6). This result does not support the assertion that local adaptation is masking relationships between cue sensitivity and range-level climatic patterns.

Discussion

In this study we compared estimates of cue sensitivity to climate conditions experienced by species across their ranges at multiple scales to test the whether spring predictability shapes cue sensitivities. Across scales, we found only weak support for the SPH, suggesting a much broader range of factors likely influence phenological cue response.

0.1 Cross-continental scale

Contrary to expectations, we found no differences in cue sensitivities between North American and European species. These results call into question recent work suggesting that European plant species successfully invade North American ecosystems because their higher reliance on forcing cues allows them to leafout earlier and gain a growth advantage over their competitors (Dawson-Glass et al., 2025). While these kinds of phenological priority effects have been documented as contributing to the success of invaders (Buonaiuto and Wolkovich, 2023; Alexander and Levine, 2019), our findings indicate that other mechanisms are likely more important for explaining the success of European woody plants in North America. Instead, this finding may help us understand why many European timber species have been successfully established in Northern America (and vice versa), without becoming aggressive on the landscape (CITE). We should note that when we subset this analysis to include only the 29 species for which we could find well-developed range maps, we did in fact observe European species to have a weaker chilling sensitivity (Figure S1, need to make it), which may further explain the pervasiveness of the idea that European invaders are successful in North America due to weaker secondary phenological cues.

Our results are less surprising when considered in light of the evolutionary relatedness of European and North American floras and increasing evidence of phylogenetic conservatism

in plant phenology. Recent studies have reported phylogenetic conservatism in phenological cue responses—suggesting related taxa will share similar sensitivities to cues, especially chilling and forcing (CITE). Given that many closely related congeners are found in both North America and Europe (CITE), we should thus a priori expect similar cues across continents. This imprint of shared evolutionary history on phenology and the shared floras across continents suggest that patterns of cue use diverged among taxa well before the current configuration of continents, and under different climate conditions than those experienced today (CITE, check Zanne or such?).

0.2 Within-continental scale

Within Europe, we found that spring predictability was associated with increased chilling and forcing sensitivity, especially when spring predictability was measured as variation in growing degree days to last frost. These patterns qualitatively match previous findings from Zohner et al. (2017), which report stronger reliance on secondary cues (i.e. chilling) in North America compared to Europe based on substantial differences in STV between the continents. But we did not find this relationship in North America, despite the continent spanning a much wider range of spring predictability than Europe. North American however had substantially weaker spring predictability (i.e., higher STV) than Europe, suggesting that the relationship may occur at certain levels of predictability. Further, our results suggest the exact metric used to approximate ‘spring predictability’ may matter, as we only found a relationship when we used a metric that estimated variability in the consistency of thermal warmth before the last frost (variability in the number of growing degree days before the last frost), a metric closely related to spring forcing cues that may be a more realistic proxy for spring predictability than STV, which measures ...

Our finding of the major difference in spring predictability between Europe and North America supports previous work (CITEZohner I think? Others?) that suggests the arrival of spring in Europe is always relatively predictable from a growing degree day perspective.

The large differences between spring predictability in Europe and North America, combined with strong signals of shared evolutionary history, suggest we should be cautious in interpreting weak relationships as evidence of a major force structuring differences in phenological cues across species. While we found marginal support for the SPH considering the climate patterns experienced by species across their ranges, our results were always weak and often depending on the metric of spring predictability: lower spring predictability was associated

with increased reliance on chilling (i.e. stronger chilling sensitivity) when using STV as a metric of predictability, and with increased reliance on photoperiod when using Stdv in GDDs to last frost as a metric for North American species, while for European species spring predictability was associated with increased reliance on chilling (i.e. stronger chilling sensitivity) when we used Stdv in GDDs to last frost as a metric. Further, all other aspects of the predictability/cue relationship either showed no relationship (i.e. forcing & photoperiod effects with STV in Europe & North America) or observed relationships contrary to the expectations of the SPH (i.e., chilling in Europe with STV and forcing in North America with Stdv. GDDs). We argue that these mixed results should not serve to refute the claims of the SPH—it is clear that spring predictability has a role to play in shaping where and how species successfully leaf out. However, our study indicates a more complicated picture of the factors that shape species’ level differences in phenological cues.

0.3 Bridging from continental to local scales

Our results—and many previous studies Zohner et al. (2017)—show high variability in cues across species with highly overlapping ranges (Make Figure S2) suggesting continental or even regional variation in climate is unlikely to explain a large portion of the observed variability, while increasing research suggests community-level processes may drive this variation. Numerous studies of species in one location—for example forest trees in one community—have found variability in cues across species (CITelaube,Flynn) of a similar scale to that seen across large scale studies (CITEZOHNER,Ettinger). Ecological theory predicts high variability in many systems where the conditions in the early season (e.g., very early spring in many temperate forests) is both more risky due to spring frost—as we have focused on here—but also rich in nutrient and light resources. Species may thus assemble each season along a gradient of growth strategies: leafout early and experience a resource-rich and low-competition, but riskier, environment, or leafout later and experience a lower resource, higher competition but less risky environment (Sakai and Larcher, 1987; Rathcke and Lacey, 1985; Lopez et al., 2008). Such a trade-off makes even more sense from a functional trait framework, which predicts early leafing species to possess traits indicative of less costly leaf and wood tissues that are easier to replace (Reich et al., 1997; Wright et al., 2004), while also being less competitive under low resource environments (and vice versa). Indeed, increasing research is finding support for such relationships between leaf out and plant traits (Wang et al., 2021; Sporbert et al., 2022), suggesting growth strategies and leafout timing within a community may be most predictive of variation in cues across species.

The importance of leafout timing within a community context further suggests we would only expect spring frost predictability to be important for a subset of species that are generally leafing out early across their communities. Thus understanding how spring frost predictability affects phenological cues may be best approached through new studies that consider both better metrics of spring predictability and community context that integrates functional traits. Research has long shown that different species have evolved multiple ways to handle spring frosts: late-leafout species effectively avoid it altogether, while early-leafout species may tolerate it through multiple pathways—building tissues that can withstand lower temperatures (through wax, trichomes etc.) or through the ability to quickly rebuild leaves. Diversity in species growth strategies and traits can lead to differences in species temporal niche, alleviating competition within a community and promoting greater species coexistence (Fargione and Tilman, 2005). These realities, alongside our findings and increasing research on woody plant phenology within communities and functional trait frameworks, suggest better predictions may come from new studies that tackle this variation, while also controlling for phylogenetic relatedness, which has been repeatedly shown to predict species’ freeze tolerance and related traits (CITES, Zanne – see who cited Zanne for some recent stuff? And Nacho should have some refs!).

At the same time, future work should include sub-species-level variation to better understand how the populations that underlie species-level variation will respond to a warming climate. Our results add to the debate about the extent to which local adaptation shapes phenological cues (Zeng and Wolkovich, 2024), and it is likely the strength of local adaptation variation among species, and phenological phases themselves (Zeng and Wolkovich, 2024; Savolainen et al., 2007). We find that species-level variation clearly swamps population-level variation in cues, but that different studies also vary substantially even after accounting for most treatment, species and population-level effects—though we found data to test this was highly limited. This suggests fully teasing out the relative effects will require more data alongside models that can fully partition variation. Given that ancestral alleles may shape current climatic sub-populations, more work on populations and sampling across species may be most useful for predicting plant responses to climate change.

1 Conclusion:

In this study we found limited support for the assertion that the predictability of spring shapes species’ relative reliance on forcing, chilling and photoperiod cues for spring phenology. Our

results suggest that climate variability may drive cue use patterns only when it is sufficiently high, like in contemporary North America. These findings indicate that future studies of phenological cue use would benefit from a holistic integration of these bio-climatic hypotheses with phylogenetic, functional trait, and climatic legacy perspectives to fully understand the evolution of phenological cues in woody plants, and how cue use patterns will impact species performance in the face of global change at across multiple spatial and temporal scales.

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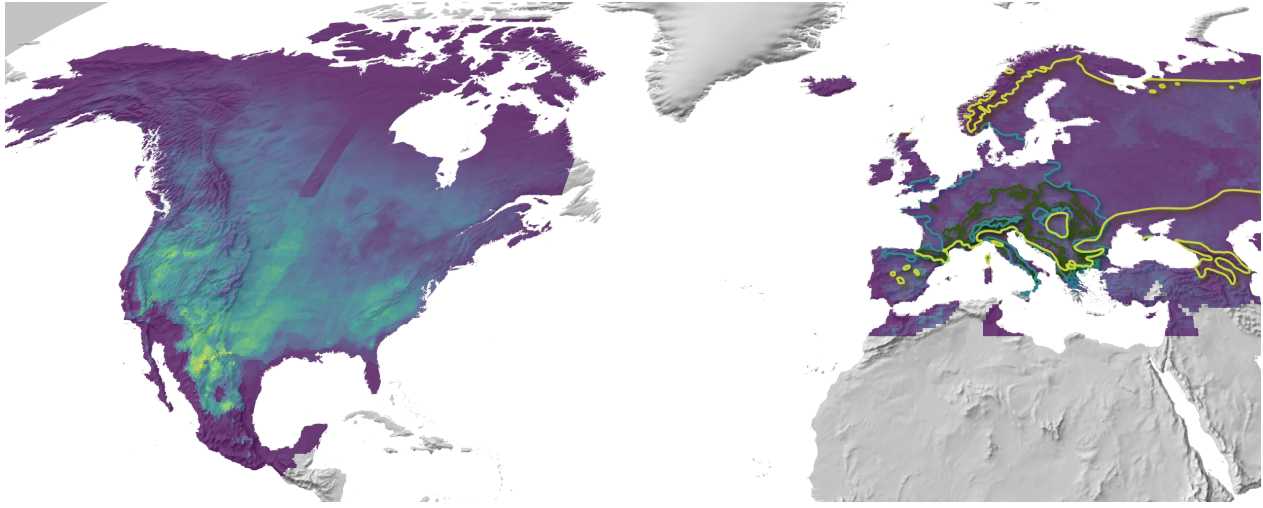


Figure 1: Contrasting patterns of spring predictability in North America and Europe. Here we represent spring predictability as the standard deviation in growing degree days to the last frost.

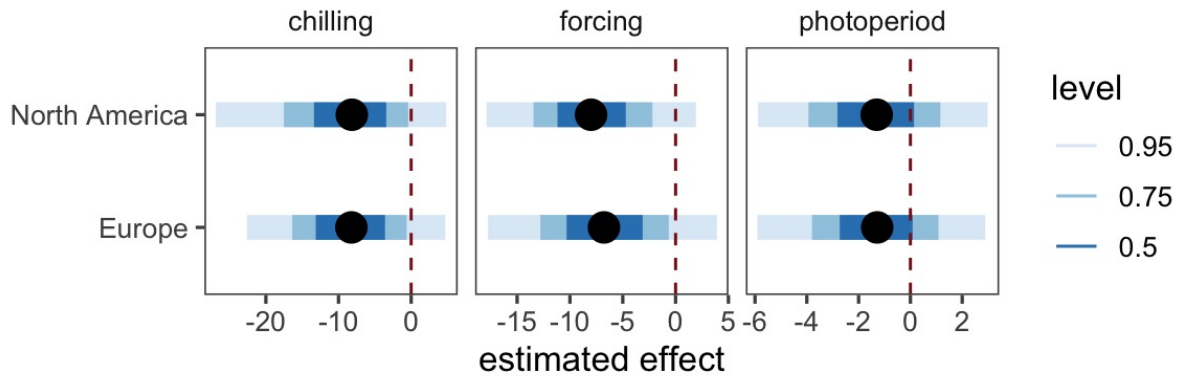


Figure 2: Estimated effect size of chilling, forcing and photoperiod on spring phenology of North American and European woody plant taxa. Point represent the mean posterior estimate and colored bars the 50, 75 and 95% credible intervals respectively.

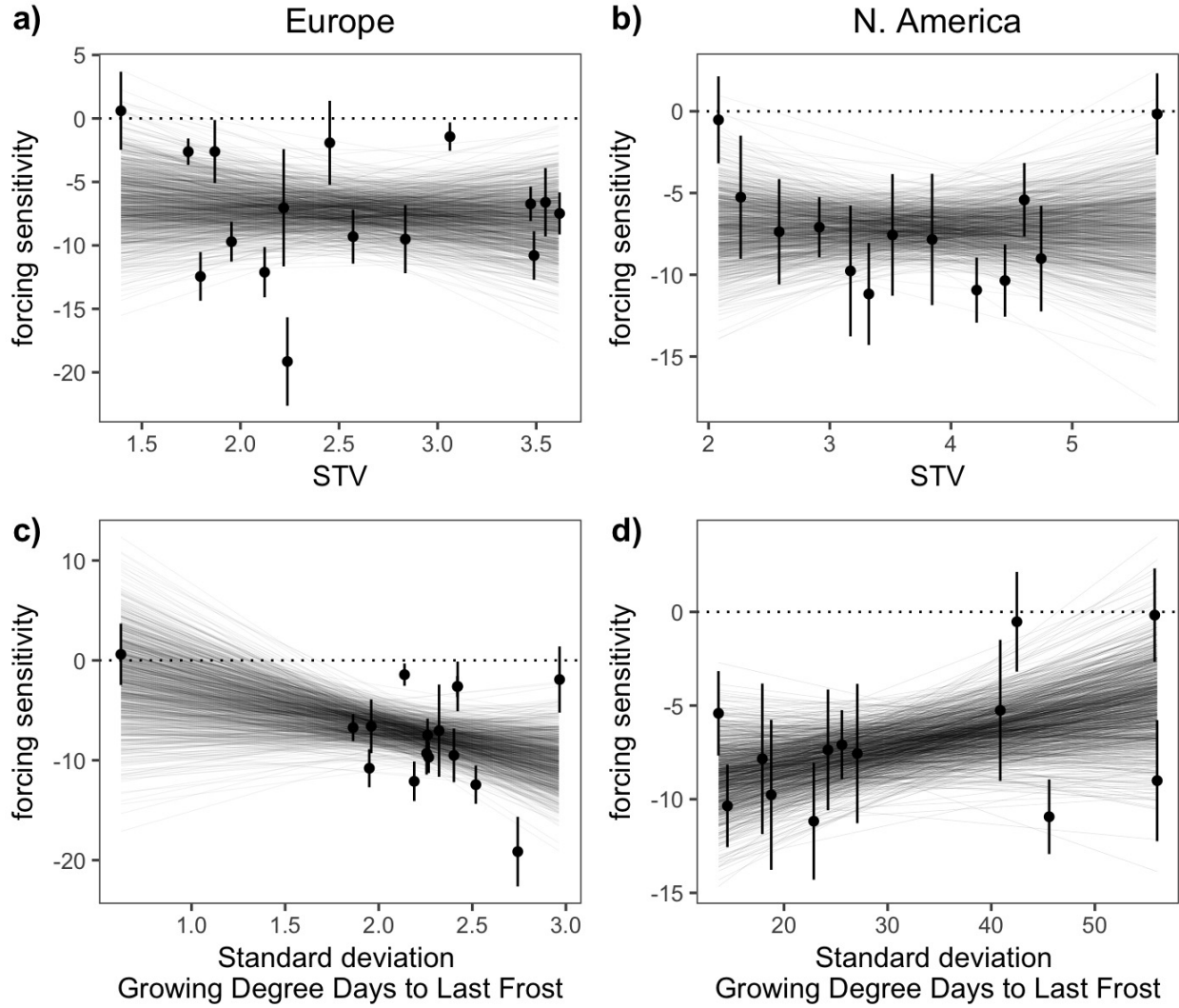


Figure 3: The relationship between species level estimates of forcing sensitivity and level of spring predictability across the species range. The points represent mean model estimates of forcing sensitivity for each species in the study, and the bars the 95% uncertainty interval. The lines represent 1000 draws from the posterior distribution capturing the relationship between STV (a,b) or the standard deviation in growing degree days to last frost (c,d) and cue sensitivity for European and North American species respectively.

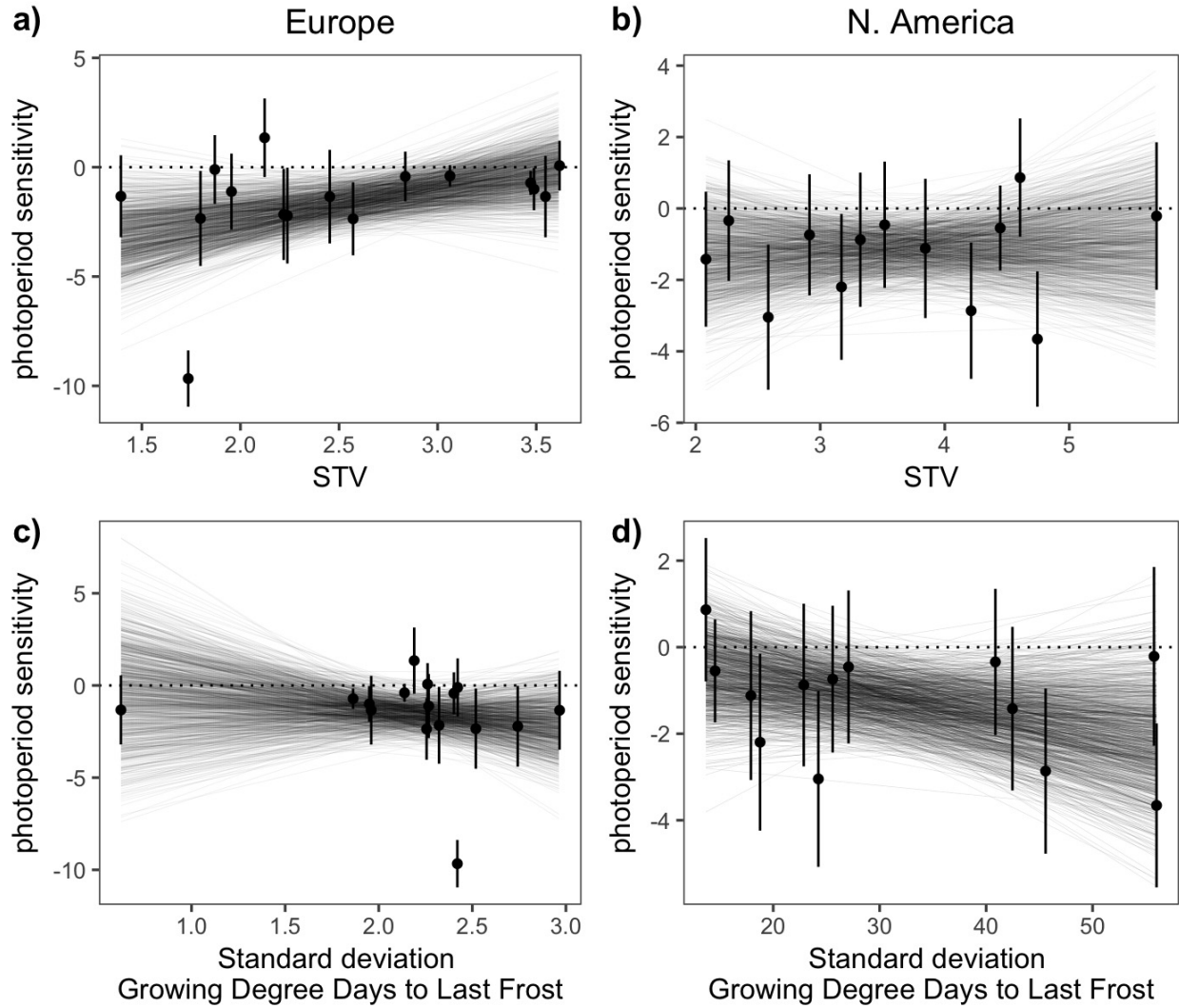


Figure 4: The relationship between species level estimates of photoperiod sensitivity and level of spring predictability across the species range. The points represent mean model estimates of photoperiod sensitivity for each species in the study, and the bars the 95% uncertainty interval. The lines represent 1000 draws from the posterior distribution capturing the relationship between STV (a,b) or the standard deviation in growing degree days to last frost (c,d) and cue sensitivity for European and North American species respectively.

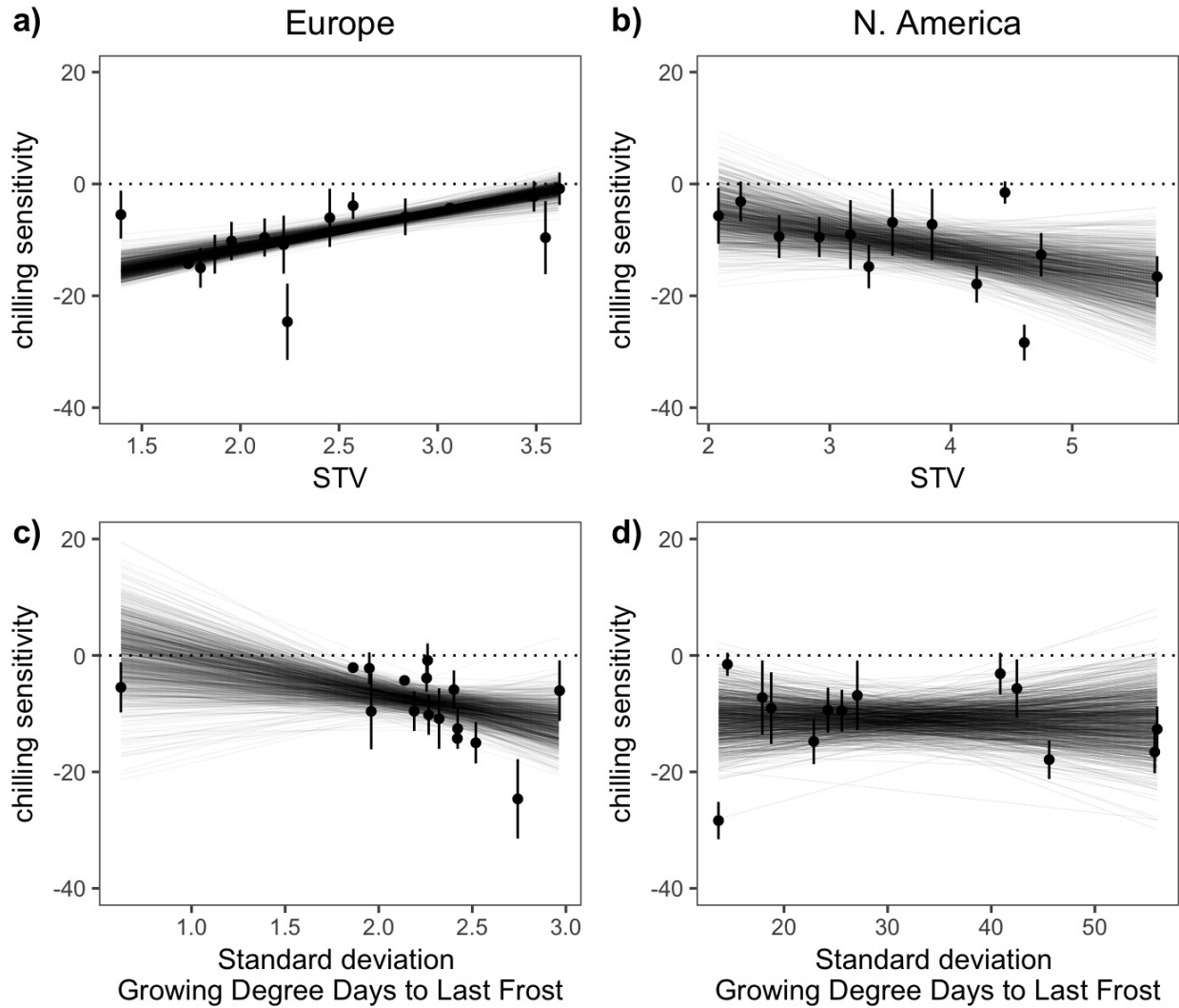


Figure 5: The relationship between species level estimates of chilling sensitivity and level of spring predictability across the species range. The points represent mean model estimates of chilling sensitivity for each species in the study, and the bars the 95% uncertainty interval. The lines represent 1000 draws from the posterior distribution capturing the relationship between STV (a,b) or the standard deviation in growing degree days to last frost (c,d) and cue sensitivity for European and North American species respectively.

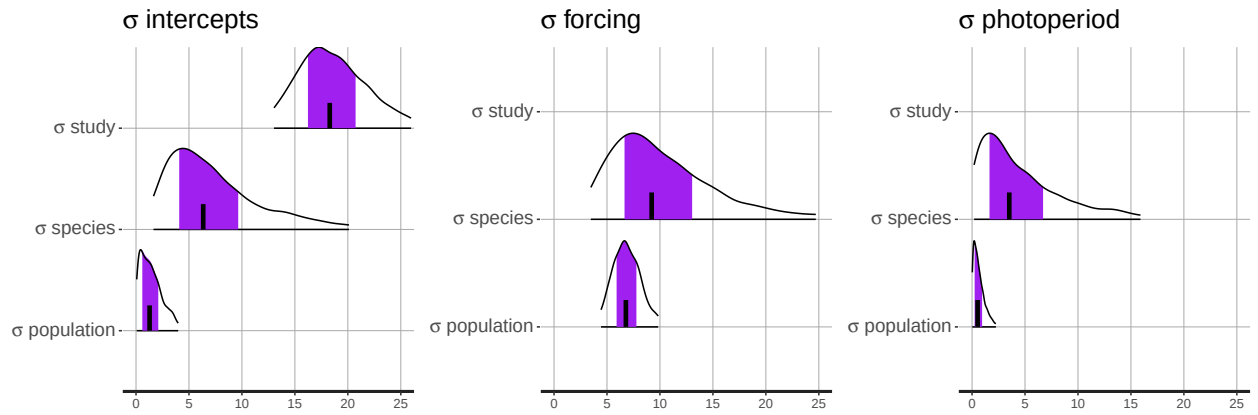


Figure 6: **Local adaptation model estimates of variation partitioning in the intercept and forcing and photoperiod predictors using the OSPREE dataset.** For both the forcing and photoperiod predictors, within species (intra-specific) variation is much smaller than across species (inter-specific) variation. Here we see that inter-specific variation exceeds intra-specific variation at the intercept-level as well but variation at the study level is largest, suggesting experimental design is driving the highest level of uncertainty.